

## Initiating Search

November 10, 2025, 2:12 PM

## • Search:

CAS Lexicon Search:

Dess-Martin periodinane - Preferred Concept

## Search Tasks

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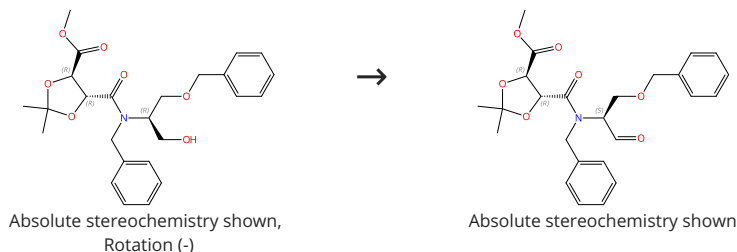
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# Reactions (50)

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## Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-10852629

Steps: 1 Yield: 100%

1.1 **Reagents:** Dess-Martin periodinane  
**Solvents:** Dichloromethane

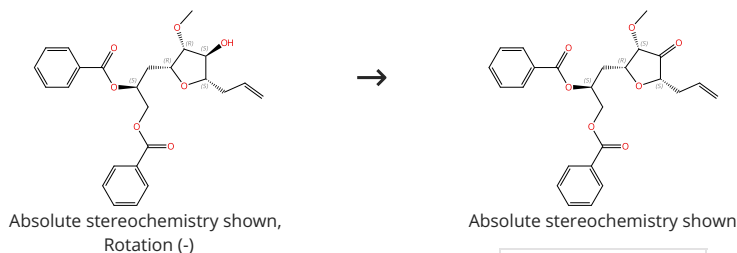
**Bicyclic compounds derived from tartaric acid and α-amino acids (BTAs): synthesis of new molecular scaffolds derived from the combination of (R,R)-tartaric acid and L-serine**

By: Cini, Nicoletta; et al

European Journal of Organic Chemistry (2002), (5), 873-880.

## Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (5)

 Suppliers (2)

31-614-CAS-41696333

Steps: 1 Yield: 100%

1.1 **Reagents:** Sodium bicarbonate, Dess-Martin periodinane  
**Solvents:** Dichloromethane; 1 h, rt

1.2 **Reagents:** Sodium thiosulfate  
**Solvents:** Water; 5 min, rt

Experimental Protocols

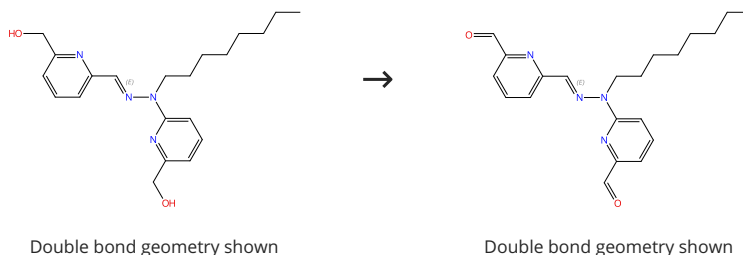
**Proprietary Synthesis of the C27-C35 Fragment of Eribulin and Its Elaboration toward the C14-C35 Subunit**

By: Kim, U. Bin; et al

Organic Process Research &amp; Development (2024), 28(7), 2832-2843.

## Scheme 3 (1 Reaction)

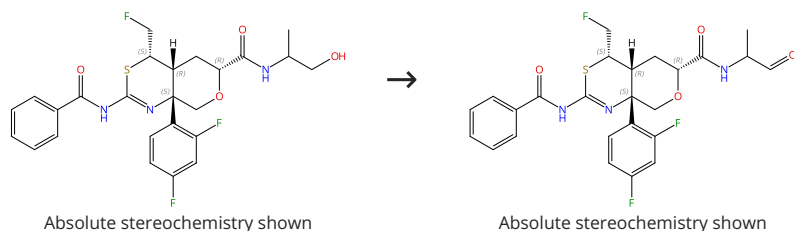
Steps: 1 Yield: 100%



|                                                                                        |                      |                                                                                                                           |
|----------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-12531468                                                                    | Steps: 1 Yield: 100% | Reversible constitutional switching between macrocycles and polymers induced by shape change in a dynamic covalent system |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 4 h, rt            |                      | By: Ulrich, Sebastien; et al                                                                                              |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Water; overnight, rt |                      | New Journal of Chemistry (2009), 33(2), 271-292.                                                                          |
| Experimental Protocols                                                                 |                      |                                                                                                                           |

## Scheme 4 (1 Reaction)

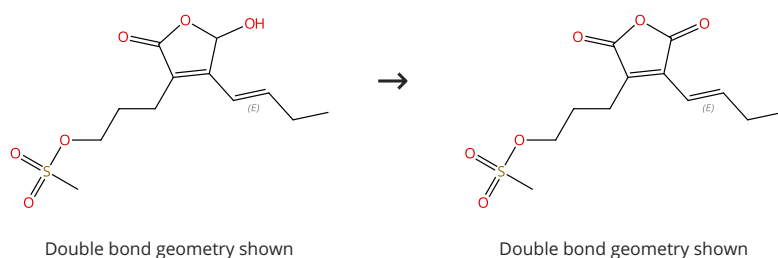
Steps: 1 Yield: 100%



|                                                                                                      |                      |                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-7188048                                                                                   | Steps: 1 Yield: 100% | Utilizing Structures of CYP2D6 and BACE1 Complexes To Reduce Risk of Drug-Drug Interactions with a Novel Series of Centrally Efficacious BACE1 Inhibitors |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 2 h, rt                          |                      | By: Brodney, Michael A.; et al                                                                                                                            |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Dichloromethane, Water; 20 min, rt |                      | Journal of Medicinal Chemistry (2015), 58(7), 3223-3252.                                                                                                  |
| Experimental Protocols                                                                               |                      |                                                                                                                                                           |

## Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



|                                                                                                                            |                      |                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 31-614-CAS-35299437                                                                                                        | Steps: 1 Yield: 100% | A new selective inhibitor for IMP-1 metallo- $\beta$ -lactamase, 3Z,5 E-octa-3,5-diene-1,3,4-tricarboxylic acid-3,4-anhydride |
| 1.1 Reagents: Sodium bicarbonate, 1,3-Dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidinone<br>Solvents: Dichloromethane; 3 h, rt |                      | By: Ikeda, Akari; et al                                                                                                       |
| 1.2 Reagents: Sodium bicarbonate<br>Solvents: Water; rt                                                                    |                      | Bioorganic & Medicinal Chemistry (2023), 78, 117109.                                                                          |
| Experimental Protocols                                                                                                     |                      |                                                                                                                               |

## Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



|                                                                    |                      |                                                                                     |
|--------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------|
| 31-480-CAS-8839513                                                 | Steps: 1 Yield: 100% | The Dess-Martin periodinane: 1,1,1-triacetoxy-1,1-dihydro-1,2-benziodoxol-3(1H)-one |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane |                      | By: Boeckman, Robert K. Jr.; et al                                                  |
| Organic Syntheses (2000), 77, 141-152.                             |                      |                                                                                     |

## Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (10)

Suppliers (5)

31-480-CAS-20303524

Steps: 1 Yield: 100%

**Rapid Optical Determination of Enantiomeric Excess, Diastereomeric Excess, and Total Concentration Using Dynamic-Covalent Assemblies: A Demonstration Using 2-Aminocyclohexanol and Chemometrics**

By: Herrera, Brenden T.; et al

Journal of the American Chemical Society (2019), 141(28), 11151-11160.

1.1 Reagents: Dess-Martin periodinane

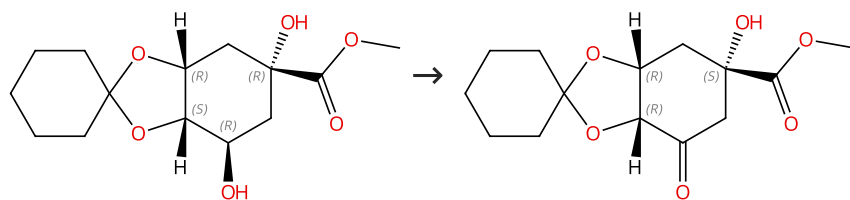
Solvents: Dichloromethane; 3 h, rt

1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate

Solvents: Water; rt

## Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-480-CAS-16246765

Steps: 1 Yield: 100%

**Synthesis, Structure, and Tandem Mass Spectrometric Characterization of the Diastereomers of Quinic Acid**

By: Deshpande, Sagar; et al

Journal of Agricultural and Food Chemistry (2016), 64(38), 7298-7306.

1.1 Reagents: Dess-Martin periodinane

Solvents: Dichloromethane; 18 h, rt

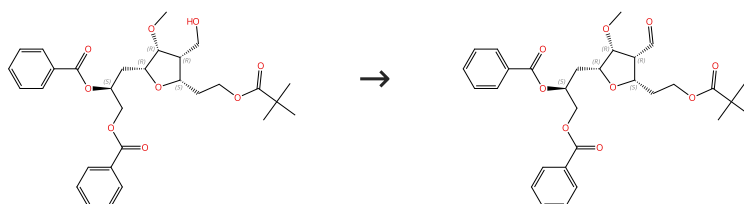
1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate

Solvents: Diethyl ether, Water

Experimental Protocols

## Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-614-CAS-41696328

Steps: 1 Yield: 100%

**Proprietary Synthesis of the C27-C35 Fragment of Eribulin and Its Elaboration toward the C14-C35 Subunit**

By: Kim, U. Bin; et al

Organic Process Research &amp; Development (2024), 28(7), 2832-2843.

1.1 Reagents: Sodium bicarbonate, Dess-Martin periodinane

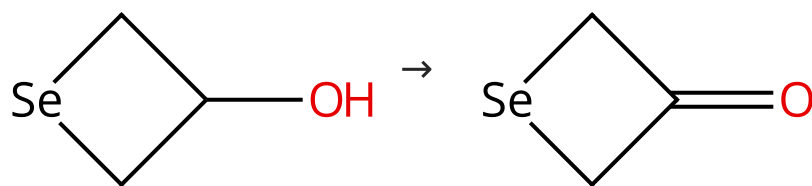
Solvents: Dichloromethane; 5 - 10 °C; 1 h, rt

1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate; rt; 30 min, rt

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Supplier (1)

31-480-CAS-9120399

Steps: 1 Yield: 100%

**The Dess-Martin periodinane: 1,1,1-triacetoxy-1,1-dihydro-1,2-benziodoxol-3(1H)-one**

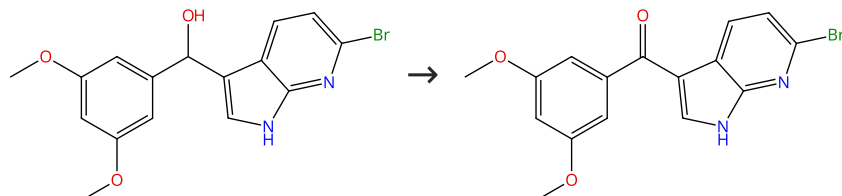
By: Boeckman, Robert K. Jr.; et al

Organic Syntheses (2000), 77, 141-152.

1.1 Reagents: Dess-Martin periodinane  
Solvents: Dichloromethane

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-23023888

Steps: 1 Yield: 100%

**Pyrrolo[2,3-b]pyridine-3-one derivatives as novel fibroblast growth factor receptor 4 inhibitors for the treatment of hepatocellular carcinoma**

By: Jin, Qiaomei; et al

Bioorganic &amp; Medicinal Chemistry (2021), 29, 115862.

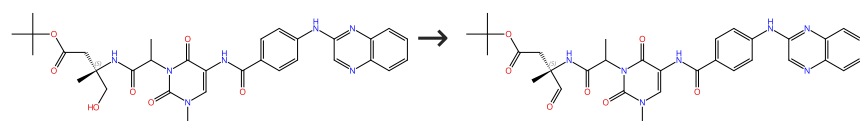
1.1 Reagents: Dess-Martin periodinane  
Solvents: Dichloromethane; 1.5 h, rt

1.2 Reagents: Sodium thiosulfate  
Solvents: Water; rt

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-480-CAS-19817684

Steps: 1 Yield: 100%

**Synthesis and stability evaluation of novel peptido mimetic Caspase-1 inhibitors for topical application**

By: Chambon, Sandrine; et al

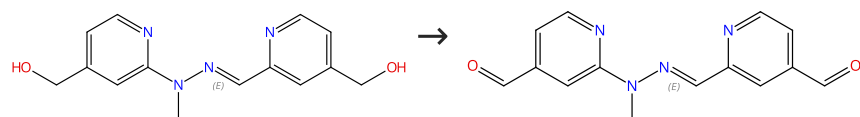
Tetrahedron (2018), 74(37), 4805-4822.

1.1 Reagents: Dess-Martin periodinane  
Solvents: Dichloromethane; 0 °C; overnight, rt

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



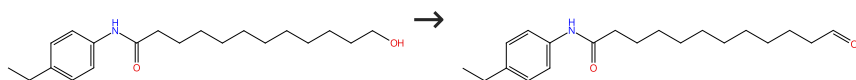
Double bond geometry shown

Double bond geometry shown

|                                                                                        |                      |                                                                                                                                  |
|----------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-5815241                                                                     | Steps: 1 Yield: 100% | <b>Reversible constitutional switching between macrocycles and polymers induced by shape change in a dynamic covalent system</b> |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 8 h, rt            |                      | By: Ulrich, Sebastien; et al                                                                                                     |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Water; overnight, rt |                      | New Journal of Chemistry (2009), 33(2), 271-292.                                                                                 |
| Experimental Protocols                                                                 |                      |                                                                                                                                  |

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



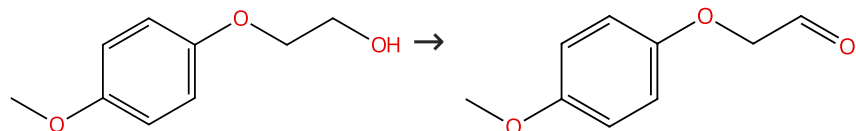
Suppliers (2)

Suppliers (2)

|                                                                                                    |                      |                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-6408948                                                                                 | Steps: 1 Yield: 100% | <b>Iodine(V) Reagents in Organic Synthesis. Part 2. Access to Complex Molecular Architectures via Dess-Martin Periodinane-Generated o-Imidoquinones</b> |
| 1.1 Reagents: 1-Hydroxy-1,2-benziodoxol-3(1 <i>H</i> )-one 1-oxide<br>Solvents: Dimethyl sulfoxide |                      | By: Nicolaou, K. C.; et al                                                                                                                              |
| Experimental Protocols                                                                             |                      | Journal of the American Chemical Society (2002), 124(10), 2221-2232.                                                                                    |

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



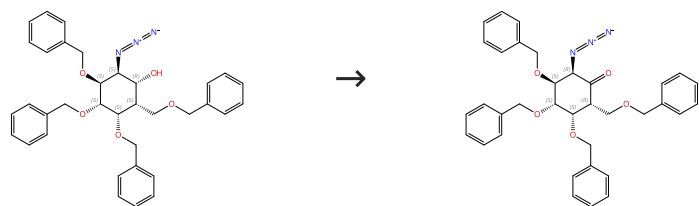
Suppliers (54)

Suppliers (27)

|                                                                               |                      |                                                                                       |
|-------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|
| 31-480-CAS-4052533                                                            | Steps: 1 Yield: 100% | <b>Direct Alkynylation of Indole and Pyrrole Hetero cycles</b>                        |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 2.5 h, rt |                      | By: Brand, Jonathan P.; et al                                                         |
| 1.2 Reagents: Sodium bicarbonate<br>Solvents: Water; rt                       |                      | Angewandte Chemie, International Edition (2009), 48(49), 9346-9349, S9346/1-S9346/61. |
| Experimental Protocols                                                        |                      |                                                                                       |

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%

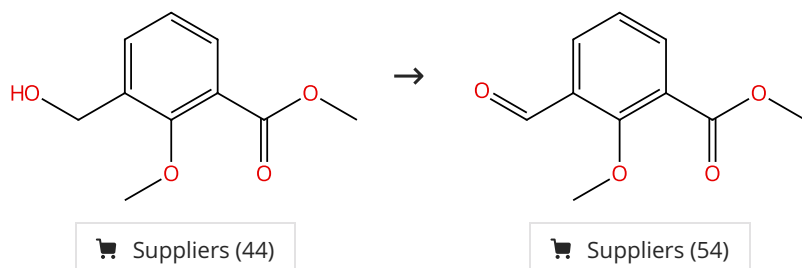
Absolute stereochemistry shown,  
Rotation (+)

Absolute stereochemistry shown

|                                                                                             |                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-24048859</b><br>Steps: 1 Yield: 100%                                          | <b><math>\alpha</math>-D-Gal-cyclophellitol cyclosulfamidate is a Michaelis complex analog that stabilizes therapeutic lysosomal <math>\alpha</math>-galactosidase A in Fabry disease</b><br><br>By: Artola, Marta; et al<br><br>Chemical Science (2019), 10(40), 9233-9243. |
| 1.1 <b>Reagents:</b> Dess-Martin periodinane<br><b>Solvents:</b> Dichloromethane; 1.5 h, rt |                                                                                                                                                                                                                                                                              |

Scheme 17 (1 Reaction)

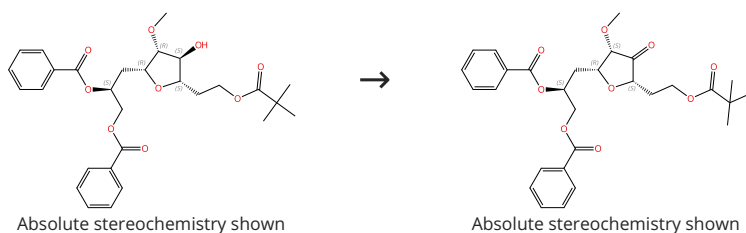
Steps: 1 Yield: 100%



|                                                                                                                                                                                                                                     |                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-480-CAS-14473594</b><br>Steps: 1 Yield: 100%                                                                                                                                                                                  | <b>Benzylic hydroxylation of aromatic compounds by P450 B M3</b><br><br>By: Neufeld, Katharina; et al<br><br>Green Chemistry (2013), 15(9), 2408-2421. |
| 1.1 <b>Reagents:</b> Dess-Martin periodinane<br><b>Solvents:</b> Dichloromethane; 30 min, rt<br><br>1.2 <b>Reagents:</b> Sodium bicarbonate, Sodium thiosulfate<br><b>Solvents:</b> Water; 45 min, rt<br><br>Experimental Protocols |                                                                                                                                                        |

Scheme 18 (1 Reaction)

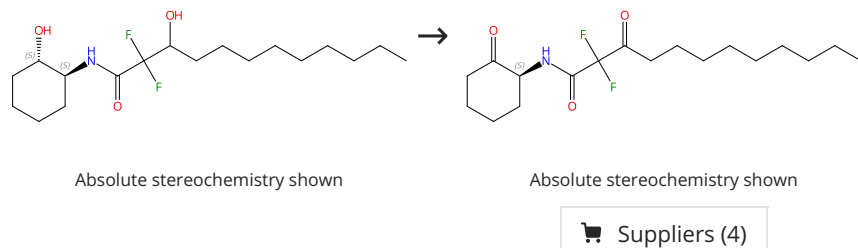
Steps: 1 Yield: 100%



|                                                                                                                                                                                                                            |                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-41696322</b><br>Steps: 1 Yield: 100%                                                                                                                                                                         | <b>Proprietary Synthesis of the C27-C35 Fragment of Eribulin and Its Elaboration toward the C14-C35 Subunit</b><br><br>By: Kim, U. Bin; et al<br><br>Organic Process Research & Development (2024), 28(7), 2832-2843. |
| 1.1 <b>Reagents:</b> Sodium bicarbonate, Dess-Martin periodinane<br><b>Solvents:</b> Dichloromethane; 3 h, rt<br><br>1.2 <b>Reagents:</b> Sodium bicarbonate, Sodium thiosulfate; 15 min, rt<br><br>Experimental Protocols |                                                                                                                                                                                                                       |

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



**Scheme 20 (1 Reaction)**

Steps: 1 Yield: 100%

The reaction scheme shows the conversion of 4-(4-(diphenylamino)phenyl)-3-yn-1-ol to 4-(4-(diphenylamino)phenyl)-3-ynal. The starting material is a diphenylamine with a 4-ethynylphenyl group, where the ethynyl group is terminated with a primary alcohol (-CH<sub>2</sub>OH). The product is the corresponding aldehyde, where the terminal hydroxyl group has been oxidized to an aldehyde (-CHO). The reaction is indicated by a right-pointing arrow.

Supplier (1)

**Scheme 21 (1 Reaction)**

Steps: 1 Yield: 100%

The reaction scheme shows the conversion of a starting material to a product. The starting material is 6-bromo-2-(hydroxymethyl)-4-methoxyphenyl 2-methoxyacetate, which is a benzodioxole derivative with a bromine atom at the 6-position, a hydroxymethyl group at the 2-position, and a methoxy group at the 4-position. The product is 6-bromo-2-(formyl)-4-methoxyphenyl 2-methoxyacetate, where the hydroxymethyl group has been oxidized to a formyl group. The reaction is indicated by a right-pointing arrow.

Suppliers (2)

**Scheme 22 (1 Reaction)**

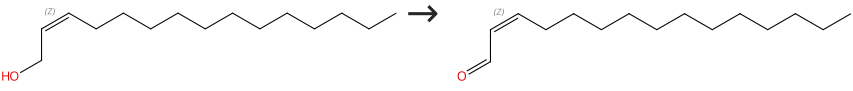
Steps: 1 Yield: 100%

Double bond geometry shown

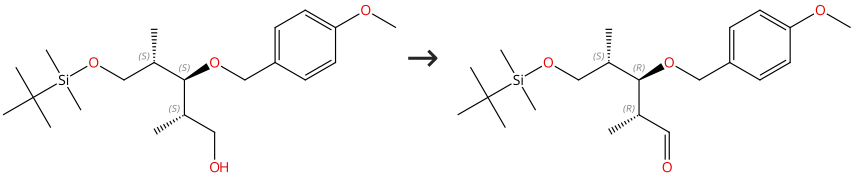
Double bond geometry shown



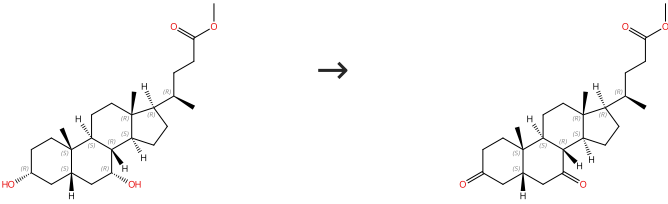
|                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-614-CAS-35299430</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Reagents:</b> Sodium bicarbonate, 1,3-Dimethyl-3,4,5,6-tetrahydro-2(1<i>H</i>)-pyrimidinone<br/><b>Solvents:</b> Dichloromethane; 3 h, rt</p> <p>1.2 <b>Reagents:</b> Sodium bicarbonate<br/><b>Solvents:</b> Water; rt</p> <p>Experimental Protocols</p> | <p><b>A new selective inhibitor for IMP-1 metallo-<math>\beta</math>-lactamase, 3Z,5E-octa-3,5-diene-1,3,4-tricarboxylic acid-3,4-anhydride</b></p> <p>By: Ikeda, Akari; et al</p> <p>Bioorganic &amp; Medicinal Chemistry (2023), 78, 117109.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Scheme 23 (1 Reaction)</b> Steps: 1 Yield: 100%</p>  <p>Double bond geometry shown      Double bond geometry shown</p> <p> Suppliers (3)</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                     |                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-7680211</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Reagents:</b> Manganese oxide (MnO<sub>2</sub>)<br/><b>Solvents:</b> Dichloromethane; 5 - 8 h; rt</p> <p>Experimental Protocols</p> | <p><b>Extended structural modulation of bio-inspired chiral lipidic alkynylcarbinols as antitumor pharmacophores</b></p> <p>By: Listunov, Dymytrii; et al</p> <p>Tetrahedron (2015), 71(41), 7920-7930.</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Scheme 24 (1 Reaction)</b> Steps: 1 Yield: 100%</p>  <p>Absolute stereochemistry shown, Rotation (-)      Absolute stereochemistry shown, Rotation (-)</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

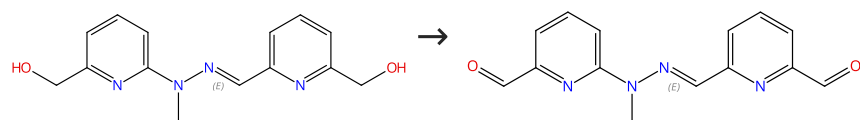
|                                                                                                                                                                                                    |                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-5134642</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Reagents:</b> Dess-Martin periodinane<br/><b>Solvents:</b> Dichloromethane; 0 °C → rt; 1 h, 0 °C</p> <p>Experimental Protocols</p> | <p><b>Probing o-Diphenylphosphinyl Benzoate (o-DPPB)-Directed C-C Bond Formation: Total Synthesis of Dictyo statin</b></p> <p>By: Wuensch, Sebastian; et al</p> <p>Chemistry - A European Journal (2015), 21(6), 2358-2363.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Scheme 25 (1 Reaction)</b> Steps: 1 Yield: 100%</p>  <p>Absolute stereochemistry shown, Rotation (+)      Absolute stereochemistry shown, Rotation (-)</p> <p> Suppliers (52)       Suppliers (11)</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                 |                                                                                    |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>31-614-CAS-34334478</b> Steps: 1 Yield: 100%                                 | <b>Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups</b> |
| 1.1 Reagents: Chromium trioxide<br>Solvents: Acetone, Water; 0 °C; 30 min, 0 °C | By: Kaspar, Miroslav; et al                                                        |
| 1.2 Reagents: Isopropanol<br>Experimental Protocols                             | Journal of Organic Chemistry (2022), 87(14), 9157-9170.                            |

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

Supplier (1)

|                                                                                        |                                                                                                                                  |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>31-480-CAS-10405011</b> Steps: 1 Yield: 100%                                        | <b>Reversible constitutional switching between macrocycles and polymers induced by shape change in a dynamic covalent system</b> |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 4 h, rt            | By: Ulrich, Sebastien; et al                                                                                                     |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Water; overnight, rt | New Journal of Chemistry (2009), 33(2), 271-292.                                                                                 |
| Experimental Protocols                                                                 |                                                                                                                                  |

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%

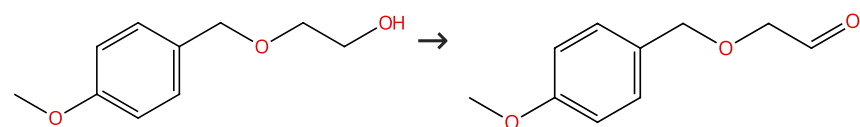
Absolute stereochemistry shown,  
Rotation (-)

Absolute stereochemistry shown

|                                                                                                       |                                                                     |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>31-614-CAS-38407793</b> Steps: 1 Yield: 100%                                                       | <b>Total Synthesis of Calcaripeptide C</b>                          |
| 1.1 Reagents: Sodium bicarbonate, Dess-Martin periodinane<br>Solvents: Dichloromethane; 0 °C; 1 h, rt | By: Patil, Shashikant; et al                                        |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Water; 0 °C                         | European Journal of Organic Chemistry (2022), 2022(47), e202201078. |
| Experimental Protocols                                                                                |                                                                     |

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



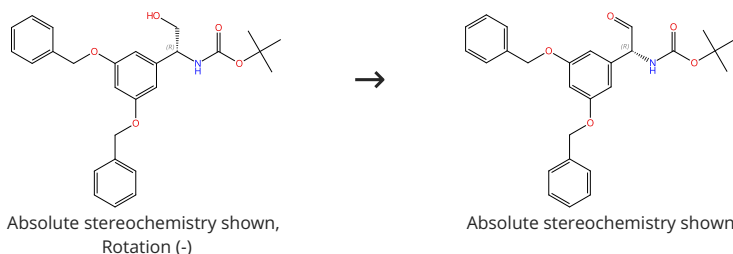
Suppliers (50)

Suppliers (28)

|                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-10691107</b> Steps: 1 Yield: 100%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; 2.5 h, rt</p> <p>1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br/>Solvents: Water; 10 min, rt</p> <p>Experimental Protocols</p> | <p><b>Copper-Catalyzed Oxy-Alkynylation of Diazo Compounds with Hypervalent Iodine Reagents</b></p> <p>By: Hari, Durga Prasad; et al</p> <p>Journal of the American Chemical Society (2016), 138(7), 2190-2193.</p> |
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Scheme 29 (1 Reaction)

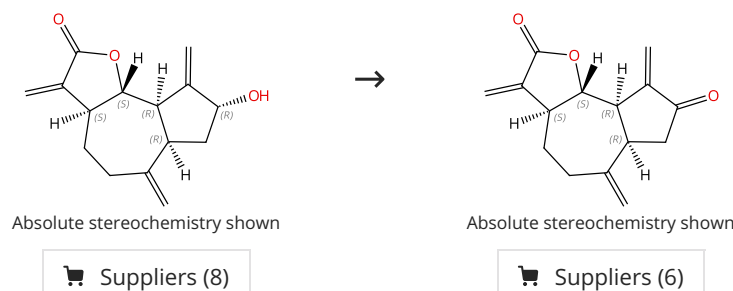
Steps: 1 Yield: 100%



|                                                                                                                                                                                                 |                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-5410936</b> Steps: 1 Yield: 100%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; rt → -5 °C; -5 °C → rt; 2 h, rt</p> <p>Experimental Protocols</p> | <p><b>Total synthesis of the antiviral peptide antibiotic feglymycin</b></p> <p>By: Dettner, Frank; et al</p> <p>Angewandte Chemie, International Edition (2009), 48(10), 1856-1861.</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 30 (1 Reaction)

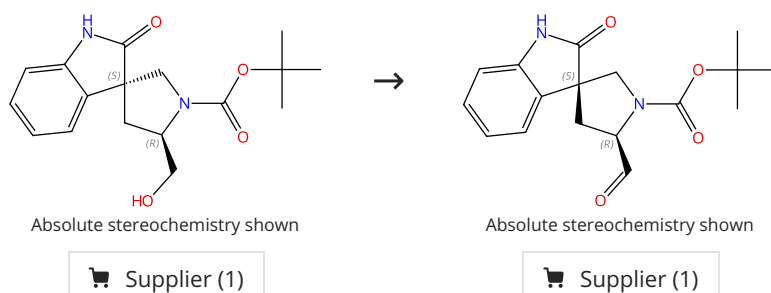
Steps: 1 Yield: 100%



|                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-23332442</b> Steps: 1 Yield: 100%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; rt; 1 h, rt</p> <p>1.2 Reagents: Sodium hydroxide<br/>Solvents: Water; rt; 15 min, rt</p> <p>Experimental Protocols</p> | <p><b>(-)-Gochnatiolide B, synthesized from dehydrocostuslactone, exhibits potent anti-bladder cancer activity in vitro and in vivo</b></p> <p>By: Chen, Yuwen; et al</p> <p>Scientific Reports (2018), 8(1), 1-11.</p> |
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Scheme 31 (1 Reaction)

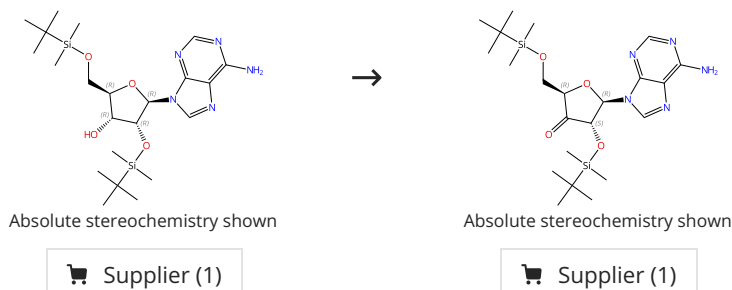
Steps: 1 Yield: 100%



|                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-4591274</b> Steps: 1 Yield: 100%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; 1 h, rt</p> <p>1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br/>Solvents: Water</p> <p>Experimental Protocols</p> | <p><b>Discovery and optimization of a novel spiropyrrolidine inhibitor of <math>\beta</math>-secretase (BACE1) through fragment-based drug design</b></p> <p>By: Efremov, Ivan V.; et al</p> <p>Journal of Medicinal Chemistry (2012), 55(21), 9069-9088.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 32 (1 Reaction)

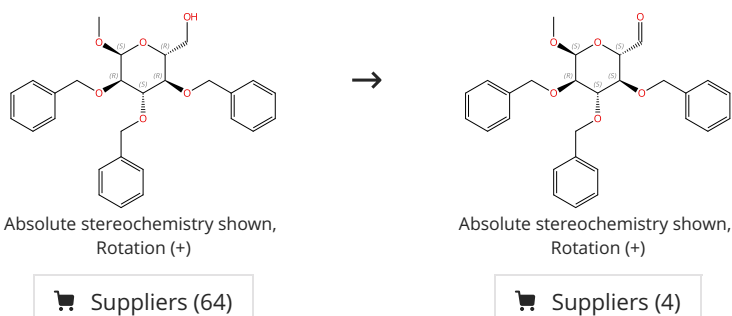
Steps: 1 Yield: 100%



|                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-2345315</b> Steps: 1 Yield: 100%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane</p> | <p><b>Nucleic acid-related compounds. 58. Periodinane oxidation, selective primary deprotection, and remarkably stereoselective reduction of tert-butyldimethylsilyl-protected ribonucleosides. Synthesis of 9-(<math>\beta</math>-D-xylofuranosyl)adenine or 3'-deuterioadenosine from adenosine</b></p> <p>By: Robins, Morris J.; et al</p> <p>Journal of Organic Chemistry (1990), 55(2), 410-12.</p> |
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Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



|                                                                                                                                              |                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-8697018</b> Steps: 1 Yield: 100%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; 30 min, rt</p> | <p><b>A mass-differentiated library strategy for identification of sugar nucleotidyltransferase activities from cell lysates</b></p> <p>By: Ko, Kwang-Seuk; et al</p> <p>Analytical Biochemistry (2013), 441(1), 8-12.</p> |
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Scheme 34 (1 Reaction)

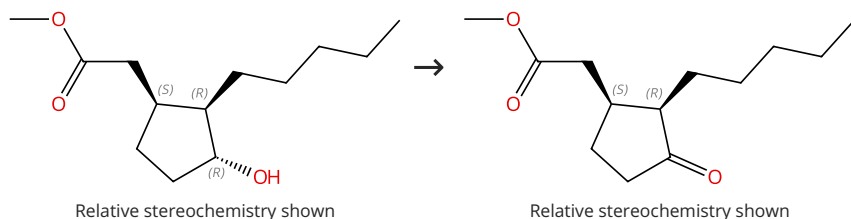
Steps: 1 Yield: 99%



|                                                                                                                                                                                         |                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-22679500</b> Steps: 1 Yield: 99%</p> <p>1.1 Reagents: Water, Dess-Martin periodinane<br/>Solvents: Dichloromethane; 0 °C; 1.5 h, 0 °C</p> <p>1.2 Reagents: Pentane</p> | <p><b>Modular Synthesis of Chiral NHC Precursors and Their Silver and Gold Complexes</b></p> <p>By: Szabo, Zita; et al</p> <p>Organometallics (2020), 39(19), 3572-3589.</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%

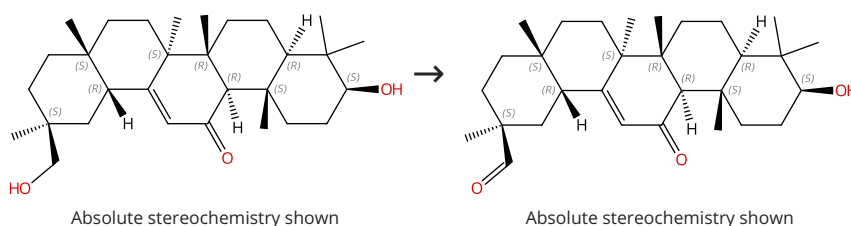


Suppliers (6)

|                                                                                                                                                                            |                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-614-CAS-40126423</b> Steps: 1 Yield: 99%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; 30 min, rt</p> <p>Experimental Protocols</p> | <p><b>Diastereoselective Hydrogenation of Tetrasubstituted Olefins using a Heterogeneous Pt-Ni Alloy Catalyst</b></p> <p>By: Swann, William A.; et al</p> <p>Angewandte Chemie, International Edition (2024), 63(19), e202317710.</p> |
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Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (5)

Suppliers (3)

|                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                      |
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| <p><b>31-480-CAS-13768958</b> Steps: 1 Yield: 99%</p> <p>1.1 Reagents: Iodobenzene diacetate<br/>Catalysts: Octahydro-2,5-dimethylpentalene-2,5-imin-7-yloxy<br/>Solvents: Dichloromethane; 1 h, rt</p> <p>1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br/>Solvents: Diethyl ether, Water; rt</p> <p>Experimental Protocols</p> | <p><b>Development of an Azanoradamantane-Type Nitroxyl Radical Catalyst for Class-Selective Oxidation of Alcohols</b></p> <p>By: Doi, Ryusuke; et al</p> <p>Journal of Organic Chemistry (2015), 80(1), 401-413.</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 37 (1 Reaction)

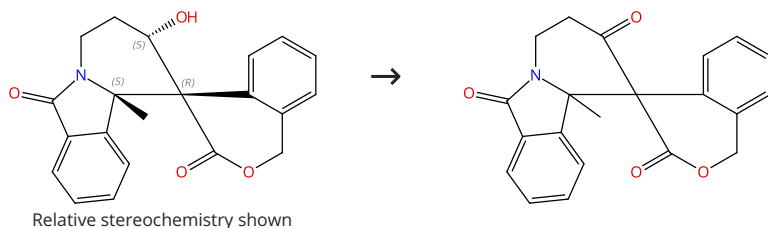
Steps: 1 Yield: 99%



|                                                                                                            |                            |                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-480-CAS-22679501</b>                                                                                 | <b>Steps: 1 Yield: 99%</b> | <b>Modular Synthesis of Chiral NHC Precursors and Their Silver and Gold Complexes</b><br><br>By: Szabo, Zita; et al<br><br>Organometallics (2020), 39(19), 3572-3589. |
| <b>1.1 Reagents:</b> Water, Dess-Martin periodinane<br><b>Solvents:</b> Dichloromethane; 0 °C; 1.5 h, 0 °C |                            |                                                                                                                                                                       |
| <b>1.2 Reagents:</b> Pentane                                                                               |                            |                                                                                                                                                                       |

Scheme 38 (1 Reaction)

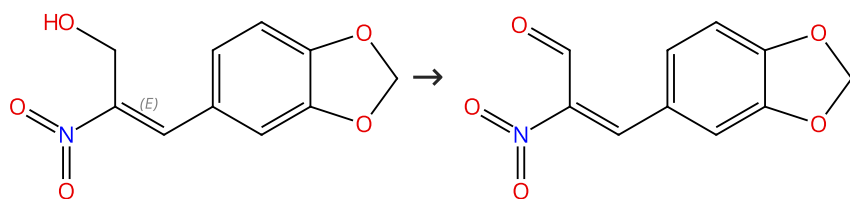
Steps: 1 Yield: 99%



|                                                                                                        |                            |                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-43333129</b>                                                                             | <b>Steps: 1 Yield: 99%</b> | <b>Regio- and Diastereoselective One-pot Double Cascade Hybrid Formation/[3+3] Spirocyclization for the Synthesis of 3-Spiropiperidines Under Phase Transfer Conditions</b><br><br>By: Di Mola, Antonia; et al<br><br>Advanced Synthesis & Catalysis (2025), 367(6), e202401230. |
| <b>1.1 Reagents:</b> 1-Piperidinocyclohexanecarbonitrile<br><b>Solvents:</b> Dichloromethane; 72 h, rt |                            |                                                                                                                                                                                                                                                                                  |
| Experimental Protocols                                                                                 |                            |                                                                                                                                                                                                                                                                                  |

Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



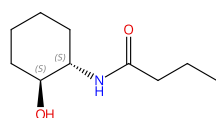
Suppliers (4)

Suppliers (2)

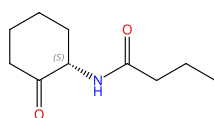
|                                                                                              |                            |                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-480-CAS-3265988</b>                                                                    | <b>Steps: 1 Yield: 99%</b> | <b>Synthesis, Structure, and E-Z Isomerization of <math>\beta</math>-(Hetero)aryl-<math>\alpha</math>-nitro-<math>\alpha,\beta</math>-enals</b><br><br>By: Martinez-Bescos, Patricia; et al<br><br>Journal of Organic Chemistry (2008), 73(10), 3745-3753. |
| <b>1.1 Reagents:</b> 2-Iodoxybenzoic acid<br><b>Solvents:</b> Ethyl acetate; 6 - 8 h, reflux |                            |                                                                                                                                                                                                                                                            |
| Experimental Protocols                                                                       |                            |                                                                                                                                                                                                                                                            |

Scheme 40 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

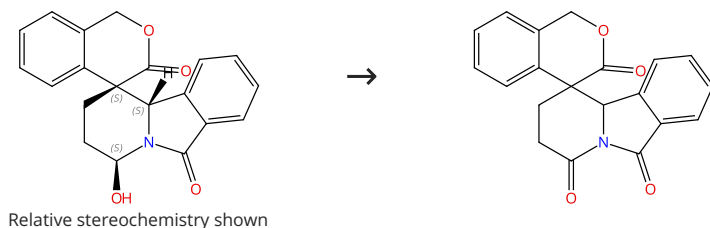


Suppliers (3)

|                                                                                                                                          |                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-480-CAS-7640746</b> Steps: 1 Yield: 99%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; 1 h, rt</p> | <p><b>Synthesis and stability of small molecule probes for Pseudo monas aeruginosa quorum sensing modulation</b></p> <p>By: Glansdorp, Freija G.; et al</p> <p>Organic &amp; Biomolecular Chemistry (2004), 2(22), 3329-3336.</p> |
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Scheme 41 (1 Reaction)

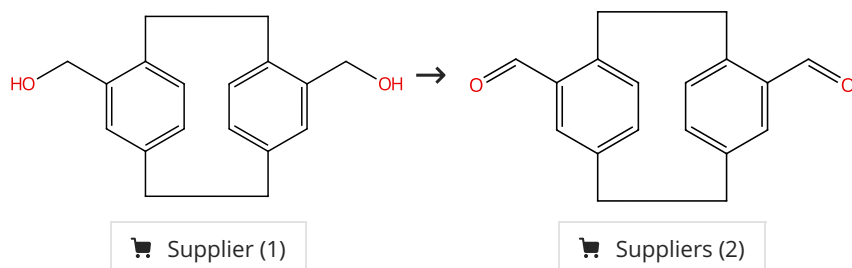
Steps: 1 Yield: 99%



|                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-614-CAS-43333132</b> Steps: 1 Yield: 99%</p> <p>1.1 Reagents: 1-Piperidinocyclohexanecarbonitrile<br/>Solvents: Dichloromethane; 72 h, rt</p> <p>Experimental Protocols</p> | <p><b>Regio- and Diastereoselective One-pot Double Cascade Hybrid Formation/[3+3] Spirocyclization for the Synthesis of 3-Spiropiperidines Under Phase Transfer Conditions</b></p> <p>By: Di Mola, Antonia; et al</p> <p>Advanced Synthesis &amp; Catalysis (2025), 367(6), e202401230.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 42 (1 Reaction)

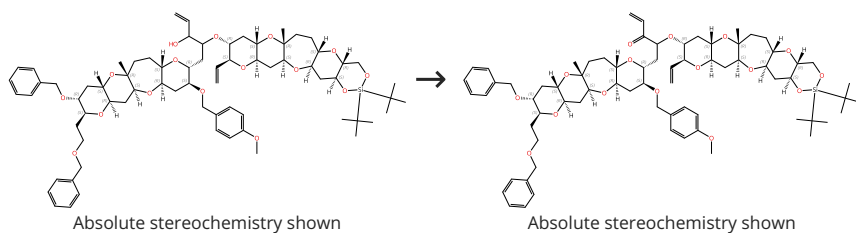
Steps: 1 Yield: 99%



|                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>31-614-CAS-44309804</b> Steps: 1 Yield: 99%</p> <p>1.1 Reagents: Dess-Martin periodinane<br/>Solvents: Dichloromethane; 30 min, 0 °C</p> <p>1.2 Reagents: Sodium bicarbonate<br/>Solvents: Water; 0 °C</p> <p>Experimental Protocols</p> | <p><b>Desymmetrization/Kinetic Resolution of Planar Chiral [2.2] Paracyclophanes by Bioinspired Peptide-Iminophosphorane Catalysis</b></p> <p>By: Wu, Zhengdong; et al</p> <p>Angewandte Chemie, International Edition (2025), 64(17), e202423702.</p> |
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Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



|                                                                                     |                     |                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-2766301                                                                  | Steps: 1 Yield: 99% | <b>Design and Synthesis of Ladder-Shaped Tetracyclic, Heptacyclic, and Decacyclic Ethers and Evaluation of the Interaction with Transmembrane Proteins</b><br>By: Torikai, Kohei; et al<br>Journal of the American Chemical Society (2008), 130(31), 10217-10226. |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 0 °C; 1.5 h, rt |                     |                                                                                                                                                                                                                                                                   |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Water             |                     |                                                                                                                                                                                                                                                                   |
| Experimental Protocols                                                              |                     |                                                                                                                                                                                                                                                                   |

Scheme 44 (1 Reaction)

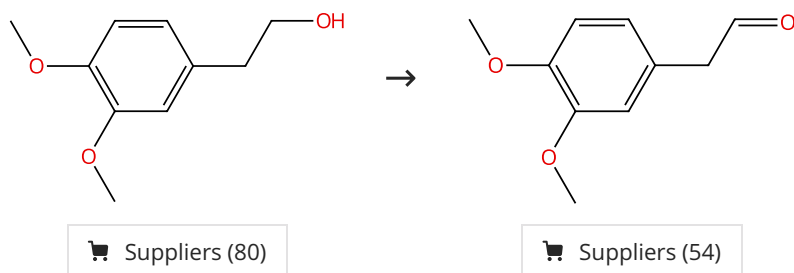
Steps: 1 Yield: 99%



|                                                                                              |                     |                                                                                                                                                               |
|----------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-22679502                                                                          | Steps: 1 Yield: 99% | <b>Modular Synthesis of Chiral NHC Precursors and Their Silver and Gold Complexes</b><br>By: Szabo, Zita; et al<br>Organometallics (2020), 39(19), 3572-3589. |
| 1.1 Reagents: Water, Dess-Martin periodinane<br>Solvents: Dichloromethane; 0 °C; 1.5 h, 0 °C |                     |                                                                                                                                                               |
| 1.2 Reagents: Pentane                                                                        |                     |                                                                                                                                                               |

Scheme 45 (1 Reaction)

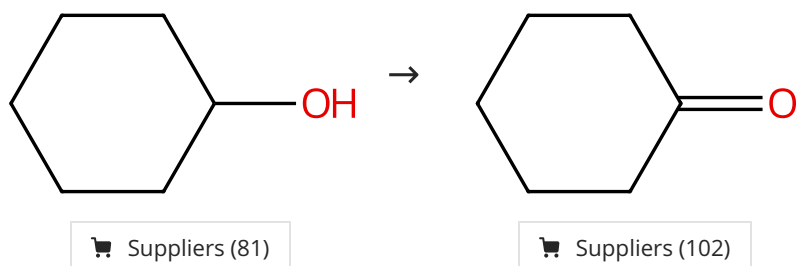
Steps: 1 Yield: 99%



|                                                                             |                     |                                                                                                                                                                                    |
|-----------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-11748703                                                         | Steps: 1 Yield: 99% | <b>Development and evaluation of new cyclooctynes for cell surface glycan imaging in cancer cells</b><br>By: Stoeckmann, Henning; et al<br>Chemical Science (2011), 2(5), 932-936. |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 1 h, rt |                     |                                                                                                                                                                                    |

Scheme 46 (1 Reaction)

Steps: 1 Yield: 99%

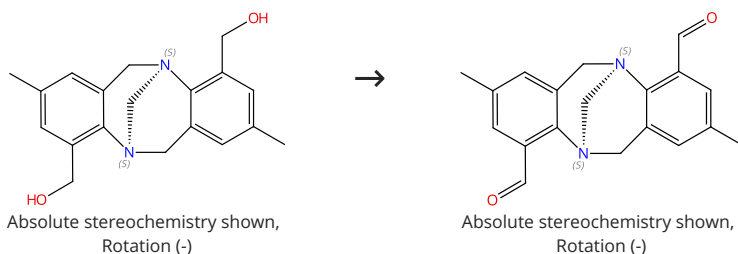




|                                             |                     |                                                                                        |
|---------------------------------------------|---------------------|----------------------------------------------------------------------------------------|
| 31-480-CAS-14763476                         | Steps: 1 Yield: 99% | <b>Dess-Martin Periodinane on Silica: Rapid Oxidation of Alcohols using Microwaves</b> |
| 1.1 Reagents: Dess-Martin periodinane; 60 s |                     | By: Heravi, Majid M.; et al                                                            |
| Experimental Protocols                      |                     | Phosphorus, Sulfur and Silicon and the Related Elements (2003), 178(4), 707-709.       |

Scheme 47 (1 Reaction)

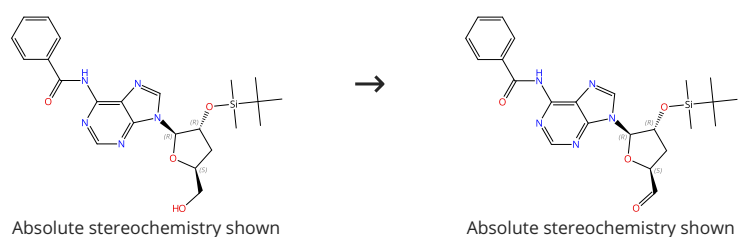
Steps: 1 Yield: 98%



|                                                                             |                     |                                                                      |
|-----------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------|
| 31-480-CAS-16663579                                                         | Steps: 1 Yield: 98% | <b>Modular Synthesis of Optically Active Troger's Base Analogues</b> |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 1 h, rt |                     | By: Kamiyama, Takuya; et al                                          |
| 1.2 Reagents: Poly(4-vinylpyridine); 20 min, rt                             |                     | ChemPlusChem (2013), 78(12), 1510-1516.                              |

Scheme 48 (1 Reaction)

Steps: 1 Yield: 98%

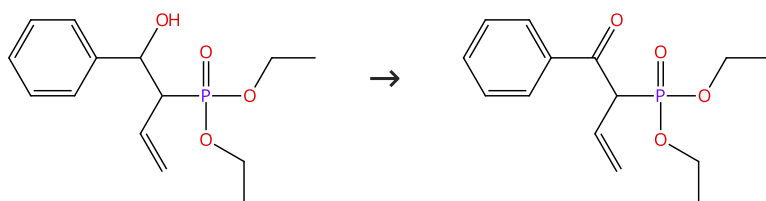


Supplier (1)

|                                                                                       |                     |                                                                                               |
|---------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------|
| 31-480-CAS-14019615                                                                   | Steps: 1 Yield: 98% | <b>A highly efficient procedure for the oxidation of the 5'-position of adenosine analogs</b> |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane; 0 °C; 1 h, 0 °C   |                     | By: Halligan, Kathleen; et al                                                                 |
| 1.2 Reagents: Sodium bicarbonate, Sodium thiosulfate<br>Solvents: Water; 10 min, 0 °C |                     | ARKIVOC (Gainesville, FL, United States) (2006), (2), 101-106.                                |

Scheme 49 (1 Reaction)

Steps: 1 Yield: 98%

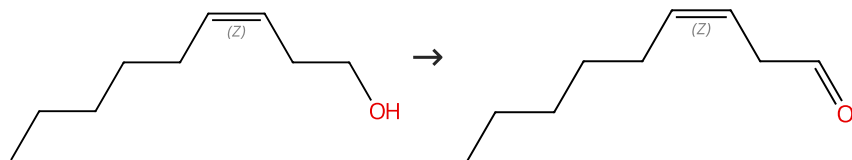


Supplier (1)

|                                                                    |                     |                                                                                                                           |
|--------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| 31-480-CAS-12101086                                                | Steps: 1 Yield: 98% | <b>Preparation of <math>\beta</math>-ketophosphonates and their vinylogues by oxidation of the corresponding alcohols</b> |
| 1.1 Reagents: Dess-Martin periodinane<br>Solvents: Dichloromethane |                     | By: Muller, Elmar L.; et al                                                                                               |
|                                                                    |                     | Organic Preparations and Procedures International (1999), 31(2), 216-220.                                                 |

## Scheme 50 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Suppliers (66)

Double bond geometry shown

Suppliers (17)

31-480-CAS-468647

Steps: 1 Yield: 98%

1.1 Reagents: Dess-Martin periodinane  
Solvents: Dichloromethane

Experimental Protocols

Clean and efficient oxidation of homoallylic and homopropargylic alcohols to  $\beta,\gamma$ -unsaturated aldehydes by the Dess-Martin periodinane

By: Wavrin, Laurence; et al

Synthesis (2002), (3), 326-330.